A robot that acts to know

Day 2 — The CORTEX architecture: one belief, many owners

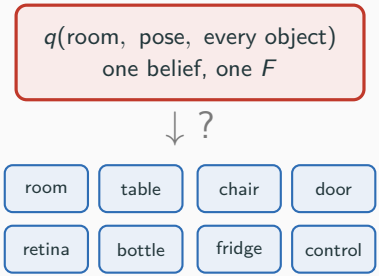
Course built on the doctoral thesis of Noé Zapata

Tuesday 1 December

Montevideo, 30 November – 4 December 2026

Who holds the belief?

What yesterday's equation did not say



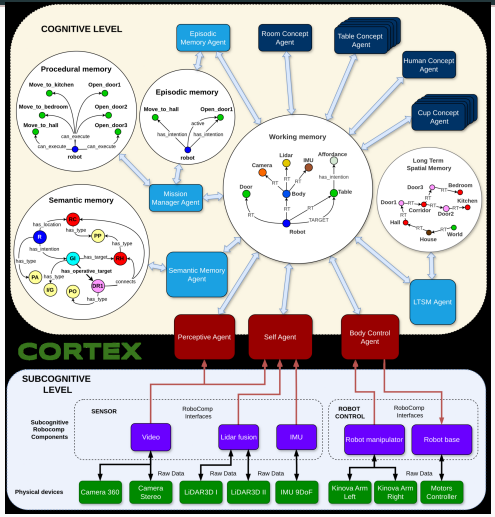
A dozen processes, sensors at 30 Hz, a robot in motion.

- **Where** is the objective computed?
- **Who** holds the beliefs?
- What happens when they must **agree** on what the world contains?

The answer is an architecture, **CORTEX**, and today's claim is that it is a factorisation of q.

source: cortex.tex l.107–109, l.377–378

CORTEX

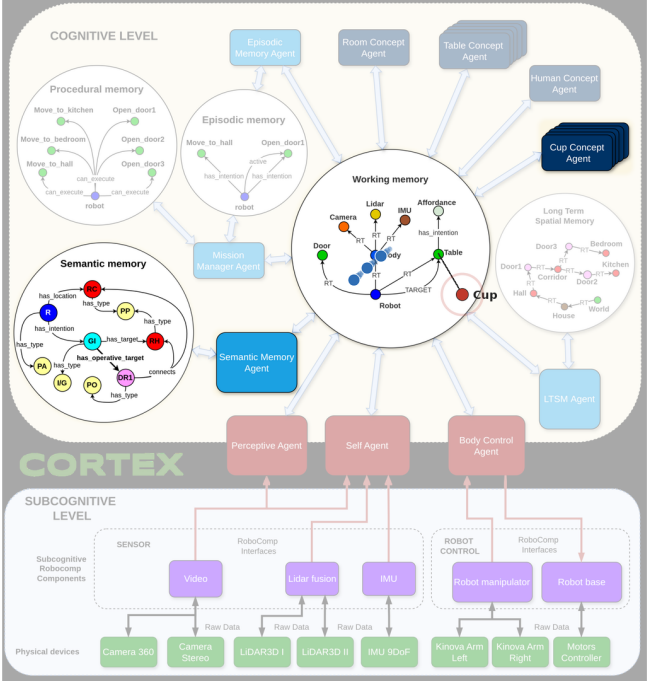


A cognitive architecture for autonomous robots: memories as graphs, and agents attached to them.

- concept agents: one class, one belief
- agents that manage the memories
- agents that reach the body
- RoboComp components

At the centre, the **working memory**: the current state, as one distributed graph. Today lives there.

source: cortex.tex l.110–113, l.154–170



CORTEX in action

1. Sensors fire
2. Two facts land on the graph
3. Concept agents read it
4. They revise their beliefs

5. Something new is born

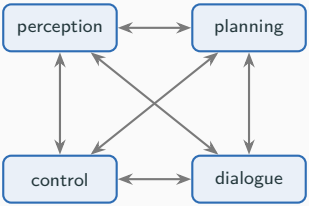
A cup node appears: a structural change. Semantic memory checks it makes sense.

6. A mission is supplied
7. An object makes an offer
8. The controller decides
9. Episodic memory keeps the record
10. The robot leaves the room

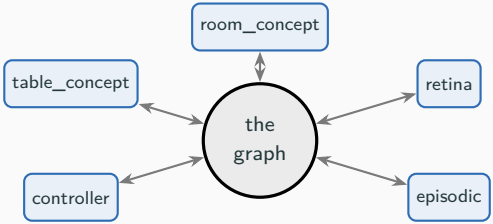
■ sensor data ● beliefs ◆ offers, actions

Schematic. On the robot all of this runs at once, at different rates.

Modules that talk, or agents that share



$n(n-1)/2$ interfaces



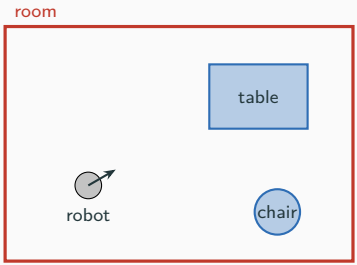
n connections

There are no agent-to-agent messages. The count is the least of it: there is one place that answers "what does the robot currently believe?"

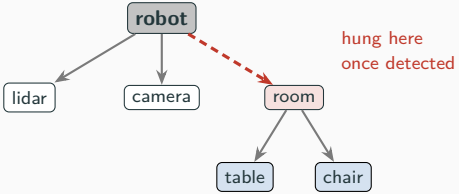
source: cortex.tex l.345–360

The shared graph

The working memory: a scene, written as a graph



what is out there



what the robot believes

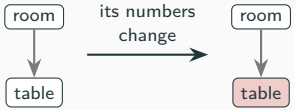
The robot is at the centre of everything. Once the room is detected, it is hung from the robot, and the objects from the room.

Where is the table, seen from the lidar? Name two nodes and ask the graph.

source: cortex.tex l.231–246

Two kinds of change

Something known has moved



The belief is revised by data.

The world contains something new

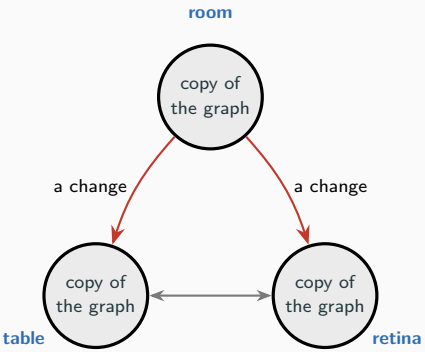


The structure of the belief itself changes.

Agents watch for one kind or the other. Tomorrow is about the second.

source: cortex.tex l.248–253, l.659–671

Everyone holds a copy, and sends corrections

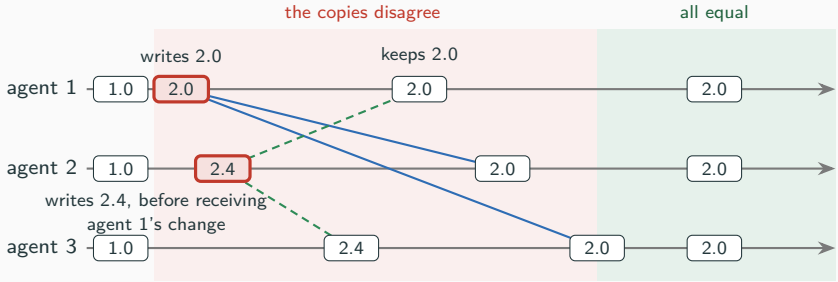


No central server, and nobody waits for a lock. Changes merge by themselves. Two consequences:

- Changes are signed by their writer, so two agents with one name corrupt the graph.
- Your copy may be a moment behind mine. Nothing may assume we see the same instant.

source: cortex.tex l.255–272

Copies may disagree for a moment, and then they agree

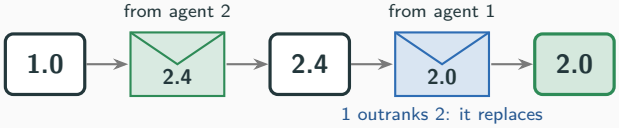


Nobody waited, and nobody locked. For a while agent 3 believes 2.4. "Eventually consistent" means only that, once the messages have arrived, every copy holds the same value.

source: cortex.tex l.255–272

Same letters, any order, same result

One copy receives agent 2's letter first



Another copy receives agent 1's letter first



Each wrote before receiving the other's change, so a fixed rule decides, the same in every copy: **the lower id wins**. A writer that had already received the other's change simply replaces it.

But the ids are arbitrary!



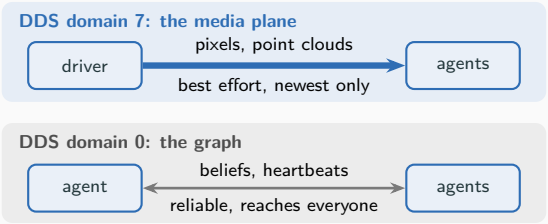
The id picks *who* wins, never *whether* the copies agree: every copy ranks the same two letters the same way.

The rule buys agreement, not the better value.

One of the two writes is always lost. That is why **each node has a single owner**: with one writer, there is nothing to collide with.

source: cortex.tex l.255–272; cortex, core/crdt/delta_crdt.h

Sensor data travels beside the graph



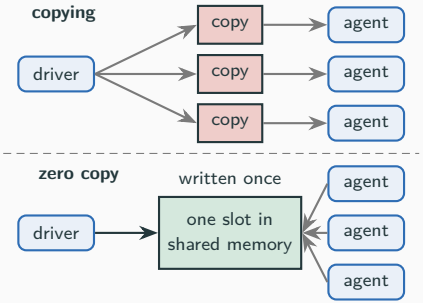
A **DDS domain** is a private channel: processes in different domains never discover each other.

Images written into the graph starve it, and the first symptom is a missed heartbeat: an agent declared dead because a camera was busy.

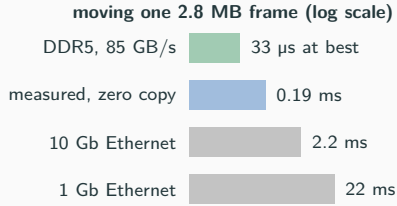
The graph holds a sensor's *existence*, and where to find its stream.

source: cortex.tex l.375–381, l.386–402, l.410–415

On one computer: pass the place, not the image



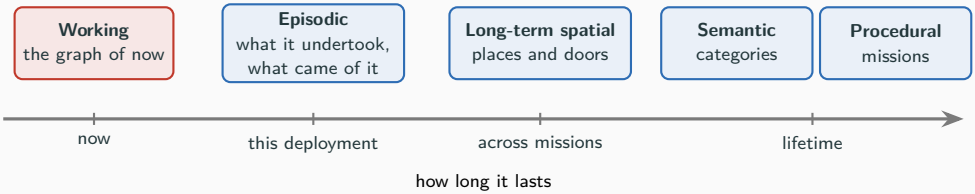
Readers look at the same bytes in place. The cost does not grow with readers.



DDR5 typically moves 80–90 GB/s. Two cameras at 30 Hz are only about 0.2 GB/s. Most of the 0.19 ms is not moving bytes.

source: cortex.tex l.400–406, l.432–435; arithmetic: 2.8 MB at each rate

Five memories, separated by timescale



Each is a distributed graph on the same substrate, with an agent that manages it. The week lives almost entirely in the red box.

source: cortex.tex l.519–537

One free energy, many owners

One big belief about everything

What objects are here?



object agents

Where am I, in what room?



room_concept

Whose mask is whose?



the matching

How do surfaces become pixels?



where they meet

Before any data, the four are independent: objects do not know where the robot is.

Once data arrive they are tangled: the scan that locates the wall also hits the refrigerator standing against it.

source: concept_agents.tex l.255–293

Not everything unknown is worth reporting

Answer it

the table's shape,
whether it is real

published on the graph,
with how sure

Allow for it

the robot's exact path,
the exact room

needed to read the data,
its doubt is carried along

Just guess it

which mask belongs
to which object

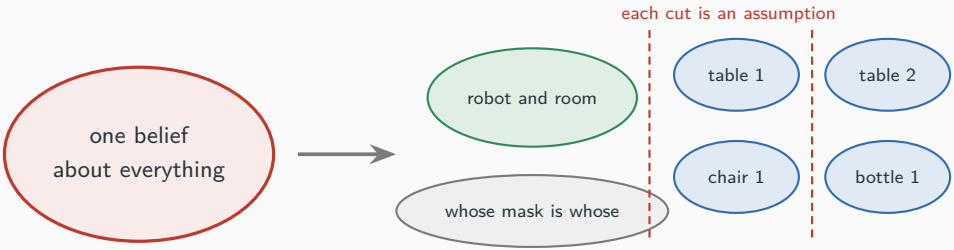
one best guess, because
trying all is impossible

The third has no principled defence, and the thesis says so.

A SLAM system would also *report* the path. That one change is what separates this design from a monolithic one.

source: concept_agents.tex l.395–420

Cut the belief into pieces, one process per piece

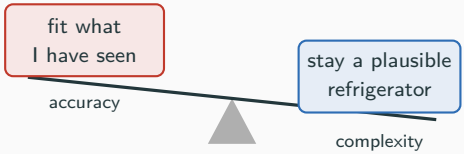


Across a cut, two beliefs no longer inform each other, except through what is published.

A process boundary is a statement about the world, not only about the software.

source: concept_agents.tex l.114–123, l.447–455

What each agent minimises: yesterday's trade, once per object



Exactly Day 1's F, for one instance:

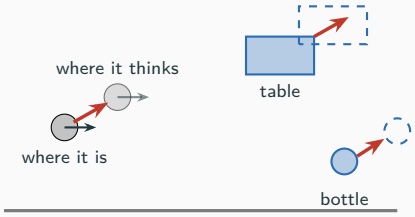
$$F_i = \text{misfit} + \text{distance from the class}$$

A refrigerator is about 0.60 ± 0.08 m deep before it is ever seen: the class does a lot of the work.

The agent takes the robot's published pose as given, and classes never add their F together.

source: concept_agents.tex l.347–351, l.496–516

The smallest scene that shows the price



the same arrow, for everything in this frame

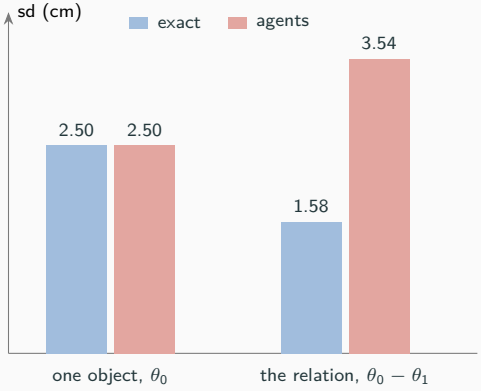
The robot places what it sees using its own pose. If the pose is off, every object in that frame is off by the same amount:

$$z_{it} = \underbrace{\theta_i}_{\text{true place}} + \underbrace{\delta_t}_{\text{shared}} + \underbrace{\varepsilon_{it}}_{\text{its own}}$$

Without a shared error, the sum of the agents' F_i is the global F. With one, dividing the work is an assumption.

source: lab 2, part C; the shared offset: concept_agents.tex l.64–70

Each belief is right. The relation between them is wrong.

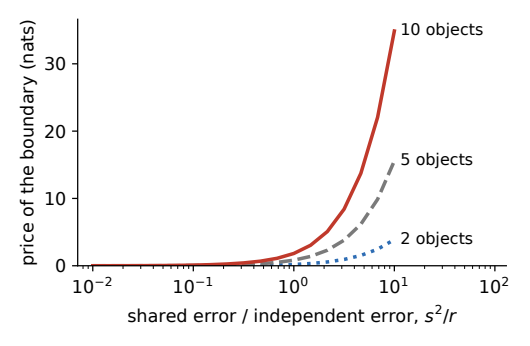


The shared error cancels in a difference. So the robot knows how far the bottle is from the table far better than it knows where either is.

Two independent beliefs on a graph cannot say that. The price is paid in relations.

source: lab 2, exercise C2 ($r = (5\,\text{cm})^2$, $s^2 = (10\,\text{cm})^2$, 20 frames, 3 objects)

The price of a boundary, in nats



How far the agents' beliefs are from the exact one, in Day 1's units:

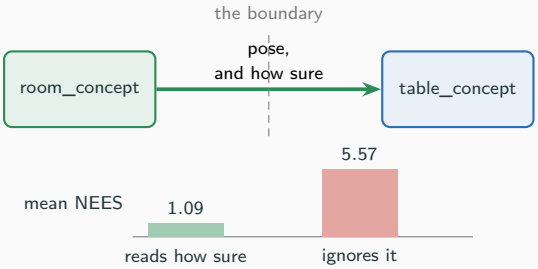
$$F[\text{agents}] - F[\text{exact}]$$

- grows with shared error and with the number of objects;
- does not fall with more data: 2.56 nats at 10, 40 and 160 frames.

Data sharpen each factor. They cannot restore a correlation the family has no place to hold.

source: lab 2, exercise C3

A mean is half a belief



The NEES compares a belief's real errors with the errors it claims: **1 is honest**.

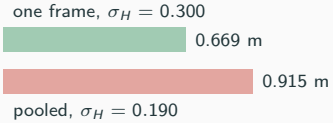
An agent that reads *how sure* the robot is stays honest. One that takes the pose as truth claims five times too small a variance.

The covariance travels with the mean, so any process can use the estimate as evidence and not as ground truth.

source: lab 2, exercise B2; concept_agents.tex l.90–95

The architecture carries its failures on its face

Pooling eight frames into one



Each frame brings its own offset, the fit has nowhere to put eight of them, so they land in the shape. More confident about a worse answer.

A cabinet labelled as a fridge



The comparison that would settle it, how well each class explains the same pixels, is never made.

source: concept_agents.tex l.73–85; agents_life_cycle.tex l.808–809

Concept agents

A concept agent is neither a detector nor a service

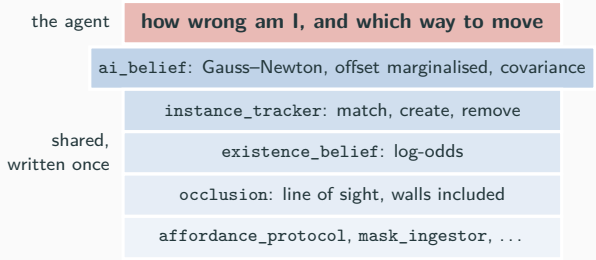


The camera only ever delivers fragments, one face at a time, never the object.

The object is a construct the agent maintains across fragments, and each cycle's belief is the next cycle's prior.

source: concept_agents.tex l.90–95, l.163–167

An agent writes two things

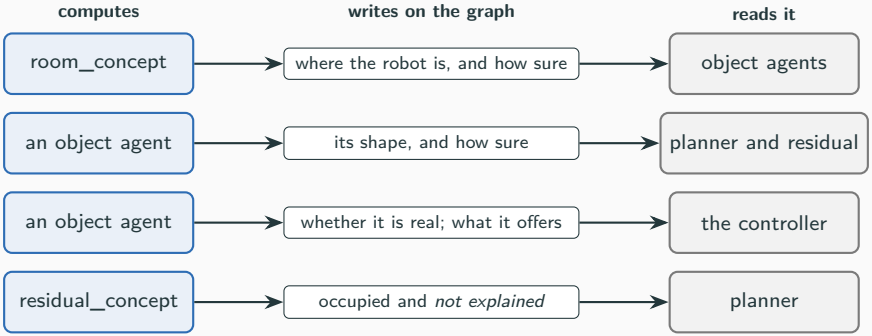


A residual and its Jacobian. All the Bayesian mathematics lives in the engine, once.

This is why a roster entry can be half a page: bottle needs little, table and refrigerator fill the template, and that is a fact about the objects.

source: concept_agents.tex l.534–537, l.136–138, l.1196–1203

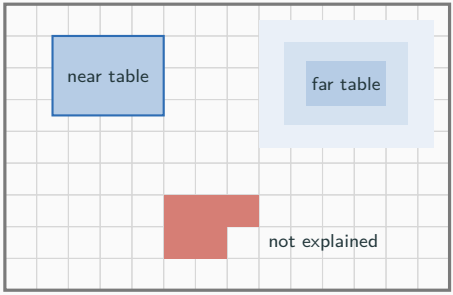
Each publishes what only it can compute



Nobody asks anybody. Everything an agent knows about another's work is on the graph.

source: concept_agents.tex l.1209–1216, l.1533–1544

residual_concept: whatever nobody else explains



a sure table claims its cells tightly; an unsure one loosely

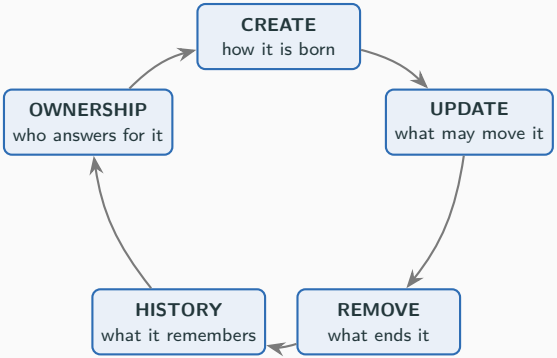
A grid of what the sensors hit, minus what every other agent claims to explain. What is left is what the robot must avoid.

Nothing has to be recognised for the robot not to hit it.

No safety margin anywhere: each claim is as wide as the claimant's own published uncertainty. That is why the others must publish it.

source: concept_agents.tex l.1497–1538

Every agent answers to a common contract



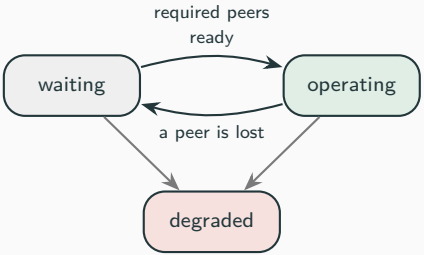
Not paperwork: this is what anything that integrates noisy fragments has to do.

A rule written in a contract is a rule a new agent cannot forget.

Tomorrow: where beliefs come from and how they end.

source: concept_agents.tex l.169–180; CONCEPT_AGENT_LIFECYCLE.md l.7–13

The factorisation is only true while its owners are alive



Specified in TLA$^+$ and model-checked: 8 safety and 4 liveness properties, no counterexample.

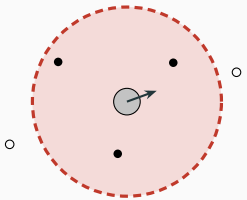
A process crashes and its node stays. It looks like any other data, and nobody is keeping it true any more.

A peer that is present is *running*, which is not the same as *able to serve*. Heartbeats are ordinary graph nodes.

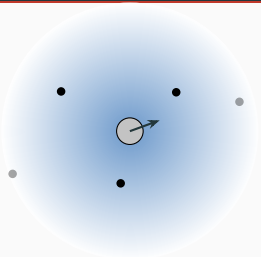
source: cortex.tex l.732–741, l.757–775, l.780–785

No thresholds

A gate is what a model looks like before the modelling is done



a gate
"ignore anything beyond 5 m"



a precision
every sighting counts, as much as it can be trusted

far away: less sure about direction

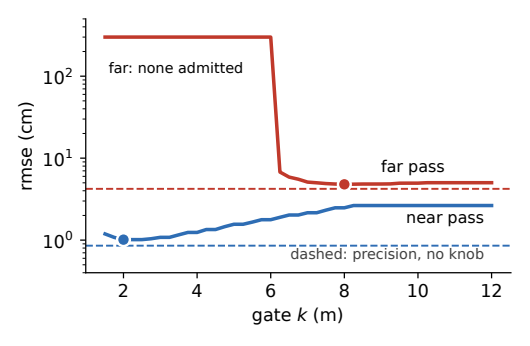
moving: the image is blurred, so a larger variance

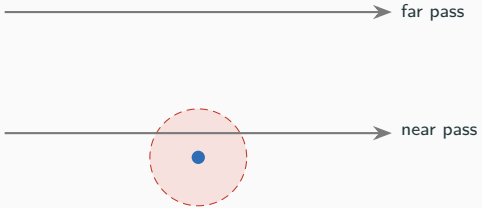
many points: they share one error, so they stop helping

No rule on top of the inference: the behaviour falls out of the model.

source: concept_agents.tex l.552–562

A gate is a parameter the scene can invalidate



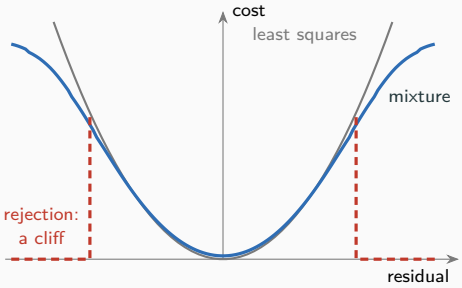


Tuned on the near pass (2 m), the gate admits nothing on the far one: 3 m of error.

The precision needs no tuning and wins on both: 0.86 against 1.01 cm, 4.22 against 4.79 cm, each against the *best* gate for that pass.

source: lab 2, part D

An outlier costs a bounded amount, and nobody rejects it



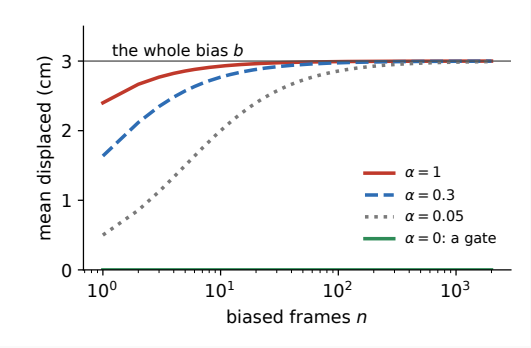
Each point has three possible explanations: the object, clutter, or a known wall.

A point that fits nothing costs a finite amount and stops pulling on the estimate.

Nobody decided to reject it. The bound is a probability in the model.

source: concept_agents.tex l.518–524

The gate that survives





a run of frames, all wrong the same way

Turning their weight α down only slows it. In the end the estimate is off by the whole bias. Only ignoring those frames stops it.

A variance handles errors that average. This one does not average.

source: agents_life_cycle.tex l.783–791; lab 2, part E

The rule is: declare it, justify it, or remove it

ignoring a bad frame: stays
no precision can bound that bias

birth after *N* sightings: must go
a count standing in for nats

waiting a few seconds for a peer: a debt
whether a peer is alive is a belief too

Measured: a test that graded itself

A chair's "can I see its yaw?" was computed from its own yaw. The yaw was about 45° wrong, so the test said no, and the wrong yaw froze.

source: agents_life_cycle.tex l.1276–1282; cortex.tex l.796–804; CONCEPT_AGENT_LIFECYCLE.md l.75–79

Lab

Lab 2: agents on a shared graph

`labs/day2.ipynb`, about two hours.

1. **The graph**: an ownership contract, and a sweep for the nodes of dead processes.
2. **Agents that only read and write**: an object agent that reads the robot's belief from the edge. Then remove the covariance and measure the NEES.
3. **One F, many owners**: when the sum of F_i is the global F; the marginal against the relation; the price of the boundary in nats.
4. **No thresholds**: a range gate against a precision, on two passes.
5. **The gate that survives**: a bias no precision can bound.

Tomorrow

Today the agents and the graph were given.
Tomorrow the beliefs on it have a life:

- birth as model comparison, not counting;
- absence as evidence, not a missed detection;
- forgetting as inferred volatility.

And the room and the robot's pose become beliefs like any other.

Reading for Day 3

Thesis chapters:

- *The Life Cycle of a Concept Agent*;
- *The Room Concept Agent and Robot Localisation*.

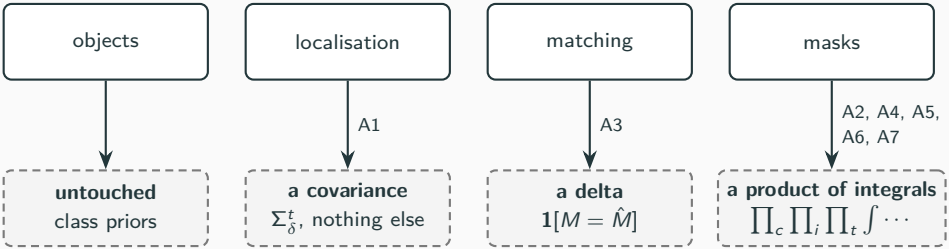
Backup: the whole scene, in causal order

objects exist and have shape	the room has walls, the robot moves, wheels report it, LiDAR hits all of it	what is visible from where	surfaces become masks
$P(\ell)\,P(\theta\|\ell)$	$P(\mathcal{R})\,P(X)\,P(\Phi\|X)\,P(L\|X,\mathcal{R},\theta)$	$P(M\|\theta,\ell,X)$	$P(Z\|\theta,M,X)$
object agents	**room_concept**	**the matching**	**where they meet**

- Objects and pose are independent *a priori*: objects do not know where the robot is.
- Once data arrive they are coupled twice: through the masks, and through the LiDAR sweep, which hits the fridge as well as the wall behind it.

source: concept_agents.tex l.255–293

Backup: from the exact posterior to what actually runs



Seven assumptions, stated one at a time, each with its justification and its cost:

A1 objects as landmarks · A2 pose error integrates per frame · A3 hard matching · A4 classes apart · A5 instances apart · A6 geometry / existence · A7 Markov time

source: concept_agents.tex l.568–613, l.626–1019

Backup: the factorised belief, and one agent's objective

$$q(\theta, \ell, M, X, \mathcal{R}) = \underbrace{q(X, \mathcal{R})}_{\text{foreign}} \; \underbrace{\mathbf{1}[M = \hat{M}]}_{\text{degenerate}} \prod_{c} \prod_{i} \underbrace{q(\theta_i)}_{\mathcal{N}(\mu_i, \Sigma_i)} \; \underbrace{q(\ell_i)}_{\text{log-odds}}$$

$$F_i = \underbrace{\mathbb{E}_q\big[-\ln P(\bar{Z}_i \mid \theta_i, \hat{X})\big]}_{\text{accuracy}} + \underbrace{D_{\mathrm{KL}}[q(\theta_i) \,\|\, P(\theta_i \mid \ell_i)]}_{\text{complexity}}$$

The localisation block is held fixed; the matching contributes nothing to F.

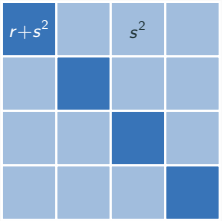
source: concept_agents.tex l.447–455, l.468–516

Backup: the shared error, as a matrix

Integrate δ_t out of $z_{it} = \theta_i + \delta_t + \varepsilon_{it}$, and the noise of the n sightings of one frame has covariance

$$C = r\, I + s^2\, \mathbf{1}\mathbf{1}^\top .$$

The agents use only its diagonal, $R = r + s^2$.



source: lab 2, part C